

Project Documentation: Amazon Fresh Analysis Dashboard

1. Executive Summary & Problem Statement

Problem Statement

Amazon Fresh is scaling rapidly, leading to a massive intake of raw transactional and user data that has become increasingly complex to manage, process, and analyze efficiently.

Proposed Solution

A structured database architecture is developed from scratch to clean, organize, and store this data. This foundational data layer powers an interactive analytics dashboard designed to:

- Identify top-performing products and inventory movement.
- Analyze demographic trends and customer shopping behaviors.
- Enable data-driven decision-making to optimize operations and drive revenue growth.

2. Dashboard Overview & Core Objectives

Primary Objectives

- **Sales Performance Tracking:** Monitor macro revenue trends and overall grocery sales volume across regional markets.
- **Customer Behavior Insights:** Track engagement, demographic profiles, and user satisfaction ratings.
- **Category Demand Monitoring:** Identify fast-moving inventory and measure category-level revenue contributions.

3. Key Performance Indicators (KPI Metrics)

- **Total Revenue:** High-level overview tracking multimillion-dollar cumulative grocery sales.
- **Customer Reach:** Total unique shopper base (1,000+ unique customers analyzed).
- **User Engagement:** Over 6,000+ customer ratings and reviews recorded to evaluate customer satisfaction.

4. Visualizations & Analytical Breakdown

Product Performance Analysis

- **Revenue Distribution by Category (Top-Right Donut Chart):** Breaks down proportional sales across various grocery categories to highlight top revenue drivers.

- **Inventory Velocity & Sales Volume (Bottom-Right Funnel Chart):** Ranks product categories by total sales volume, highlighting items with the highest inventory turnover and demand.

Audience & Demographic Profiling

- **Gender & Age Breakdown:** Displays customer distribution by age brackets and gender identities to define the core shopper persona.
- **Customer Satisfaction & Category Distribution (Left Chart):** Maps customer distribution across product categories against overall customer satisfaction ratings.

Regional Segmentation & Interactive Slicers

- **Dynamic State Slicer (Left Filter Panel):** Allows stakeholders to dynamically filter the entire dashboard by individual states (such as California, Florida, Texas) to assess localized market health and high-profit regional opportunities.

5. Summary & Key Takeaways

The Amazon Fresh Analysis Dashboard delivers end-to-end visibility from high-level enterprise metrics down to granular regional and product performance. By centralizing raw transaction metrics into an interactive interface, leadership can rapidly pinpoint revenue drivers, optimize inventory placement, and target marketing efforts effectively.

SQL Overview

This project manages the database for **Amazon Fresh**, an online grocery platform. It organizes data across customers, products, suppliers, customer orders, and product reviews to maintain data integrity and generate actionable business metrics.

Key Strengths in Your SQL Implementation

- **Analytical Ranking:** Window functions like `RANK() OVER (...)` are implemented properly to sort customers by total spend.
- **Group Filtering:** Effective use of `GROUP BY` and `HAVING` filters out aggregated metrics accurately, such as isolating customers who spent over 5,000 or products with average ratings above 4.
- **Data Integrity Checks:** Well-defined constraint rules ensure review scores stay between 1 and 5, and customer membership defaults cleanly to "No".

Logic Issues and Necessary Corrections

Customer Table Setup

- Adding new constraints or columns directly to an existing table can fail if default names or empty values conflict.
- *Fix:* Recreate the table with explicit primary key, not-null age check (greater than 18), and unique name constraints.

Date Format Syntax

- Writing dates without quotes (like 2024-01-01) causes SQL to perform mathematical subtraction instead of treating it as a calendar date.

- *Fix:* Always enclose date literals in quotes, such as '2024-01-01'.

Product Sales Ranking

- Grouping by order ID and customer ID fails to track which specific items generated revenue.
- *Fix:* Join order_details with products and aggregate by ProductID.

Order Revenue Calculation

- Summing order amounts directly from the main orders table duplicates values when joined with item details.
- *Fix:* Calculate revenue on the line-item level using SUM(Quantity * UnitPrice) from the order_details table.

Database Normalization (3NF)

- Creating a new table with hardcoded manual records does not systematically normalize category data.
- *Fix:* Create an auto-incrementing categories lookup table and link its foreign key inside the products table.

Subqueries for Top Products and Inactive Customers

- Using scalar subqueries inside the select list fails to aggregate product revenue correctly, and checking OrderID IS NULL on the orders table yields no results because primary keys cannot be null.
- *Fix:* Aggregate sales inside a subquery within the FROM clause for top products, and use WHERE CustomerID NOT IN (...) to locate inactive customers.

Prime Member and Category Demand Insights

- Counting the membership column without filtering counts non-members too, and counting product IDs measures inventory rather than actual customer purchases.
- *Fix:* Add WHERE PrimeMember = 'Yes' for city counts, and join order_details with products to measure real category purchase volume.